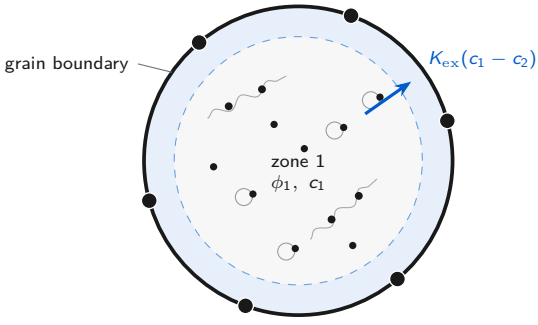
Five metal particles (FMP) in UO_2

Elisa Cappellari
November 4, 2026

- black: model of V. Bellani
- blue: new or changed equation or assumption (EC November 2026)

Two particle populations: two zones



Intragranular particles $N_{\mathrm{intra}}, C^{\mathrm{intra}}, R_{\mathrm{intra}}$

- sites: dislocations ($f_d \rho_d / \lambda$) and intragranular bubbles ($f_b N_b$)
- fed by the metal dissolved in zone 1, $c_1 = C_{m1}/\phi_1$

Grain-boundary particles $N_{\mathrm{GB}}, C^{\mathrm{GB}}, R_{\mathrm{GB}}$

- sites: the boundary surface, $3/(2a\lambda^2)$
- fed by the metal dissolved in zone 2, $c_2 = C_{m2}/\phi_2$

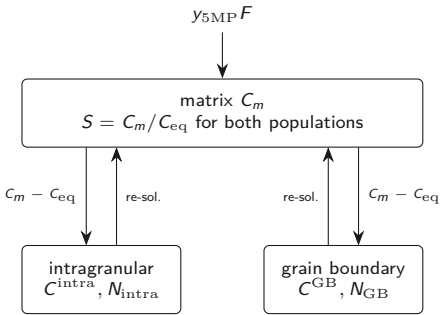
Processes

- production $\phi_k P$ in each zone; nucleation (CNT); diffusion-limited growth
- re-solution, back to the zone of the particles
- exchange between the zones, $K_{ex}(c_1 - c_2)$

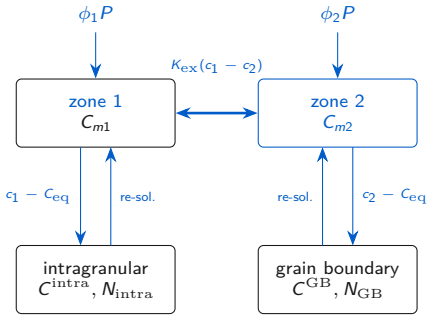
V. Bellani: one matrix C_m feeds both populations with the same supersaturation, with common parameters $R_{\mathrm{GB}} = R_{\mathrm{intra}}$.

Metal balances: one matrix vs two zones

V. Bellani (thesis Fig. 3.1, eqs. 3.5–3.7)



Two zones (EC November 2026)



Production

Lumped yield from the PRODHEL composition:

$$\dot{C}_{5\mathrm{MP}} = y_{5\mathrm{MP}} F, \qquad y_{5\mathrm{MP}} = 0.578 \text{ at/fiss} \qquad \text{[VB 3.1, 3.3]}$$

Now: ENDF/B-VIII.0 yields with the fission fractions of an LWR cycle

$$\frac{dC_{\mathrm{FMP}}}{dt} = \sum_X y_X F, \quad X = \mathrm{Mo}, \mathrm{Tc}, \mathrm{Ru}, \mathrm{Rh}, \mathrm{Pd} \qquad \textbf{EC Nov. 2026}\text{, in place of [VB 3.1]}$$

y_{Mo}	y_{Tc}	y_{Ru}	y_{Rh}	y_{Pd}	total
0.2370	0.0614	0.1442	0.0469	0.0771	0.5666 at/fiss

Grain-boundary zone

Zone 2 = shell of thickness δ inside each spherical grain of radius a, so that every boundary has a layer δ on both sides; zone 1 = grain interior.

$$\phi_2 = 1 - \left(1 - \frac{\delta}{a}\right)^3 \approx \frac{3\delta}{a} \qquad \phi_1 = 1 - \phi_2, \qquad \delta = 100 \text{ nm}$$

EC Nov. 2026

$$P_k = \phi_k \, \Delta C_{\text{prod}} \qquad c_k = \frac{C_{mk}}{\phi_k} \qquad k = 1, 2$$

- δ = half-width of the grain-boundary regions analysed by TEM in the calibration (thesis Figs. 5.3–5.16)

Equilibrium concentration and supersaturation

$$C_{\mathrm{eq}}(T) = C_{\mathrm{eq,ref}} \exp\left[-\frac{H_{\mathrm{sol}}^{\mathrm{eff}}}{k_B}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right], \qquad T_{\mathrm{ref}} = 2173.15\ \mathrm{K} \qquad \text{[VB 3.10, 3.11]}$$

$$C_{\mathrm{eq,ref}} = x_{\mathrm{eq,ref}} \frac{4}{a_{\mathrm{UO_2}}^3} = 6 \times 10^{-5} \cdot 2.437 \times 10^{28} = 1.462 \times 10^{24}\ \mathrm{at/m^3}, \qquad H_{\mathrm{sol}}^{\mathrm{eff}} = 1.14\ \mathrm{eV}$$

[VB 3.12–3.15]

Mo solubility in UO_2 at 1900 °C (Kleykamp), $a_{\mathrm{UO_2}} = 547.5$ pm, DFT solution energy of Mo (Brillant et al.).

Supersaturation after the production of the time step:

$$S = \frac{C_m^n + \Delta C_{\mathrm{prod}}}{C_{\mathrm{eq}}(T)} \qquad \text{[VB 4.2, 4.3]}$$

$$S_k = \frac{C_{mk}^n + \phi_k \Delta C_{\mathrm{prod}}}{\phi_k\, C_{\mathrm{eq}}(T)} = \frac{c_k^*}{C_{\mathrm{eq}}} \qquad \textbf{EC Nov. 2026}\text{, in place of [VB 3.16, 4.3]}$$

$$\Delta\mu_k = k_B T \ln S_k \quad (S_k > 1;\ \text{no nucleation for } S_k \le 1) \qquad \text{[VB 3.17]}$$

Nucleation: classical nucleation theory

Heterogeneous nucleation, capillarity approximation, j = intra, GB with S of the zone of the population:

$$\Delta G_j^* = \frac{B_j}{T^2(\ln S_k)^2}, \qquad B_j = \frac{16\pi\gamma_j^3\Omega_{5\mathrm{MP}}^2\Phi_j}{3k_B^2} \text{ (effective, calibrated)} \qquad \text{[VB 3.20, 3.21]}$$

$$k_{\mathrm{nucl},j} = k_{\mathrm{nucl},0,j}\exp\left(-\frac{\Delta G_j^*}{k_B T}\right), \qquad k_{\mathrm{nucl},0,j} = 1\ \mathrm{s}^{-1} \qquad \text{[VB 3.22]}$$

$$\frac{dN_j}{dt} = k_{\mathrm{nucl},j}\max\left(N_{\mathrm{site},j} - N_j, 0\right) \qquad \text{[VB 3.9]}$$

$B_{\mathrm{intra}} \neq B_{\mathrm{GB}}$, calibrated separately. V. Bellani: $B_{\mathrm{intra}} = B_{\mathrm{GB}} = B_{\mathrm{ref}} s_B$, $s_{B,\mathrm{intra}} = s_{B,\mathrm{GB}}$ [VB 6.2, 6.4].

Nucleation sites

Intragranular: dislocations (Veshchunov and Shestak) and intragranular fission gas bubbles

$$\rho_d = A\, b^n f_T, \qquad f_T = A_\infty + \frac{1 - A_\infty}{1 + \exp[(T - T_c)/\Delta T]} \qquad \text{[VB 3.23, 3.24]}$$

$$A = 6.545 \times 10^{12},\ n = 1.151,\ A_\infty = 0.608,\ T_c = 1109\ \text{K},\ \Delta T = 25.8\ \text{K},\ b \text{ in } \text{MWd/kgUO}_2 \qquad \text{[VB 3.25]}$$

$$N_{\text{site,intra}} = \phi_1 \left(f_d \frac{\rho_d}{\lambda} + f_b N_{b,\text{intra}} \right), \qquad f_d = 0.67,\ f_b = 0.33,\ \lambda = 15\ \text{nm}$$

EC Nov. 2026, in place of [VB 3.29]

Grain boundary: the boundary surface, area per unit volume $3/(2a)$ (each boundary shared by two grains), one site every λ^2

$$N_{\text{site,GB}} = \frac{A_{\text{GB}}}{V} \frac{1}{\lambda^2} = \frac{3}{2a\lambda^2} \qquad \text{[VB 3.31–3.33]}$$

Metal taken by nucleation

Atoms per nucleus: critical size of CNT, at least a dimer (Ru clusters in UO_2)

$$n_{\mathrm{nucl},j} = \max\left(2, \frac{2\Delta G_j^*}{\Delta\mu}\right), \qquad \Delta C_{\mathrm{nucl},j} = n_{\mathrm{nucl},j}\,\Delta N_j$$ [VB 3.34–3.36, 4.7, 4.8]

V. Bellani: the two requests are limited together by the dissolved metal, in proportion

$$\text{if } \Delta C_{\mathrm{nucl,intra}} + \Delta C_{\mathrm{nucl,GB}} > C_m^{\mathrm{avail}}: \quad f_{\mathrm{mass}} = \frac{C_m^{\mathrm{avail}}}{\Delta C_{\mathrm{nucl,intra}} + \Delta C_{\mathrm{nucl,GB}}}, \quad \Delta C_{\mathrm{nucl},j}^{\mathrm{act}} = f_{\mathrm{mass}}\Delta C_{\mathrm{nucl},j}$$

[VB 4.9–4.11]

Now: each population takes the atoms of its own zone ($k = 1$ for intra, $k = 2$ for GB)

$$\Delta C_{\mathrm{nucl},j}^{\mathrm{act}} = \min\left(n_{\mathrm{nucl},j}\Delta N_j,\ C_{mk}^n + \phi_k \Delta C_{\mathrm{prod}}\right)$$ **EC Nov. 2026**, in place of [VB 4.9–4.11]

$$\Delta N_j^{\mathrm{act}} = \frac{\Delta C_{\mathrm{nucl},j}^{\mathrm{act}}}{n_{\mathrm{nucl},j}}, \qquad N_j^{n+1} = N_j^n + \Delta N_j^{\mathrm{act}}$$ [VB 4.12]

Growth (precipitation)

Diffusion-limited capture by spherical particles, $4\pi DR$ per particle (Ham; Brailsford et al.)

$$k_j = D_j\, 4\pi N_j R_j, \qquad D_j = k_{0,j} \exp\left(-\frac{\Delta G_j}{k_B T}\right), \qquad \Delta G_{\mathrm{intra}} = \Delta G_{\mathrm{GB}} = 1 \text{ eV} \qquad \text{[VB 3.38, 3.39]}$$

evaluated with the state at the beginning of the time step (no iteration on $R \to k \to C$) [VB 4.13].

$$J_{\mathrm{prec},j} = k_j \max\left(C_m - C_{\mathrm{eq}}, 0\right) \qquad \text{[VB 3.37]}$$

$J_{\mathrm{prec,intra}} = k_{\mathrm{intra}}\,(c_1 - C_{\mathrm{eq}}), \qquad J_{\mathrm{prec,GB}} = k_{\mathrm{GB}}\,(c_2 - C_{\mathrm{eq}})$ **EC Nov. 2026**, in place of [VB 3.37]

each active only if its zone is supersaturated after production and nucleation, $c_k^* > C_{\mathrm{eq}}$ (V. Bellani: both active if $C_m^* > C_{\mathrm{eq}}$ [VB 4.19]). No thermal dissolution [VB Sec. 3.5].

$k_{0,\mathrm{intra}} \neq k_{0,\mathrm{GB}}$, calibrated separately: $D_{\mathrm{GB}}/D_{\mathrm{intra}} = 3.06$. V. Bellani: common k_0, $s_{k0,\mathrm{intra}} = s_{k0,\mathrm{GB}}$ [VB 6.5].

Exchange between the zones

Flux from zone 1 to zone 2 per unit fuel volume: $J_{\mathrm{ex}} = K_{\mathrm{ex}}(c_1 - c_2)$, two resistances in series: grain interior $\rightarrow$ interface (grain-boundary sink strength with the intragranular particles as internal sinks), interface $\rightarrow$ middle of zone 2 (distance $\delta/2$, area $3/a$ per unit volume).

$$K_{\mathrm{ex}} = \frac{D_{\mathrm{intra}}}{1/k_{\mathrm{GB}}^2 + a\delta/6}, \qquad k_{\mathrm{GB}}^2 = \max\left(k_1^2 \frac{F}{1-F}, \frac{15}{a^2}\right), \qquad F = \frac{3}{z}\left(\coth z - \frac{1}{z}\right), \quad z = k_1 a$$

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$$k_1^2 = 4\pi \frac{N_{\mathrm{intra}}}{\phi_1} R_{\mathrm{intra}} \quad \text{(sink strength of the intragranular particles in zone 1)}$$

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Equivalent Fick form: area $\times$ D / distance

$$K_{\mathrm{ex}} = \frac{3}{a}\frac{D_{\mathrm{intra}}}{L_1 + \delta/2}, \qquad L_1 = \frac{3}{a k_{\mathrm{GB}}^2} = \begin{cases} a/5 = 2.5\ \mu\mathrm{m} & \text{no particles (Booth sphere)} \\ \approx 1/k_1 & \text{particles dominate, } z \gg 1 \end{cases}$$

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$1/k_1$ = distance travelled by a dissolved atom before the capture by a particle. r000: $L_1 = 107$ nm at 14 and 42 nm at 54.1 MWd/kgUO$_2$, against $\delta/2 = 50$ nm. No new parameter.

Exchange: derivation

1. Sphere of radius a, uniform source S, particles as uniform sinks k_1^2, interface concentration c_i, $u = c - c_i$, quasi-steady state:

$$D\,\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{du}{dr}\right) - Dk_1^2 u + S = 0, \quad u'(0) = 0,\ u(a) = 0$$

$$u(r) = \frac{S}{Dk_1^2}\left[1 - \frac{a\sinh(k_1 r)}{r\sinh(k_1 a)}\right]$$

2. Mean, with $\int_0^a r\sinh(k_1 r)\,dr = \frac{a\cosh z}{k_1} - \frac{\sinh z}{k_1^2}$, $z = k_1 a$:

$$\bar{u} = \frac{3}{a^3}\int_0^a u\,r^2\,dr = \frac{S}{Dk_1^2}(1 - F), \quad F = \frac{3}{z}\left(\coth z - \frac{1}{z}\right)$$

3. The source goes to the particles or across $r = a$: $S = Dk_1^2\bar{u} + Dk_{\mathrm{GB}}^2\bar{u}$

$$k_{\mathrm{GB}}^2 = \frac{k_1^2}{1-F} - k_1^2 = k_1^2\frac{F}{1-F}$$

4. Limits: $z \to 0$: $\coth z = 1/z + z/3 - z^3/45$, $F = 1 - z^2/15$, $k_{\mathrm{GB}}^2 \to 15/a^2$ (Booth sphere, $\bar{u} = Sa^2/15D$); $z \gg 1$: $F \approx 3/z$, $k_{\mathrm{GB}}^2 \to 3k_1/a$.

5. Interface $\to$ middle of zone 2, area $4\pi a^2/(\frac{4}{3}\pi a^3) = 3/a$:

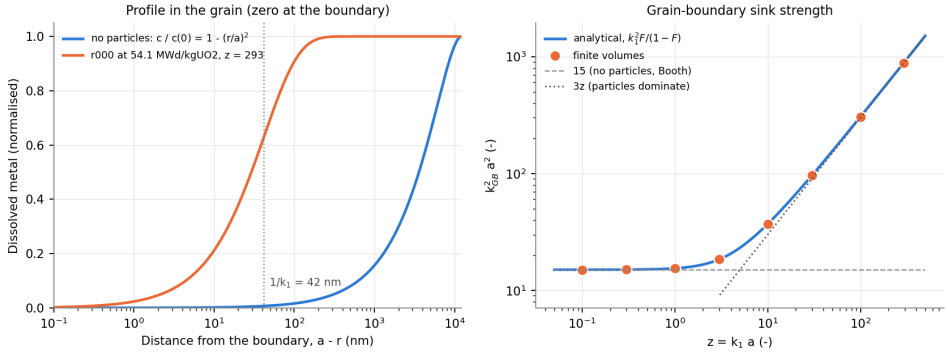
$$J = \frac{3}{a}\,\frac{D(c_i - c_2)}{\delta/2} = \frac{6D}{a\delta}(c_i - c_2)$$

6. Series: $c_1 - c_i = J/(Dk_{\mathrm{GB}}^2)$, $c_i - c_2 = Ja\delta/(6D)$, summing:

$$J = \frac{D(c_1 - c_2)}{1/k_{\mathrm{GB}}^2 + a\delta/6} = K_{\mathrm{ex}}(c_1 - c_2)$$

Exact: steps 1–4, for any c_i and with the C_{eq} term (only S changes).
Approximate: zone 2 well mixed with its mean at $\delta/2$; radius a instead of $a - \delta$ (0.8%); D_{intra} also in zone 2; quasi-steady state at each step. Code block marked [TBC]. Brailsford and Bullough (1981); Booth (1957).

Exchange: numerical check



- left: profile in the grain, without particles and with the particles of r000 at 54.1 MWd/kgUO$_2$ ($z = 293$): only the metal produced in the last $\sim$40 nm reaches the boundary
- right: $k_{\mathrm{GB}}^2 a^2$ from a finite-volume solution of step 1 vs the expression of step 3, agreement within 6×10^{-8} for $z = 0.1$–293, and the two limits of step 4 (`check_exchange.py`)

Irradiation-induced re-solution

First order in the precipitated inventory, same coefficient for the two populations

$$J_{\mathrm{res},j} = k_{\mathrm{res}}\, C^{j} \qquad \text{[VB 3.40]}$$

From the shrinkage of the FMP under ion irradiation (Sakib et al. 2025): $R_1/R_0 = 0.80/0.91$ between 0.30 and 1.75 dpa, $n \propto R^3$, $\dot{D}_{\mathrm{ref}} = 1$ dpa/day

$$k_{\mathrm{res}} = -\frac{3}{\Delta t}\ln\frac{R_1}{R_0}, \quad \Delta t = \frac{\Delta D}{\dot{D}_{\mathrm{ref}}} \;\Rightarrow\; k_{\mathrm{res,ref}} = 3.085\times 10^{-6}\ \mathrm{s}^{-1} \qquad \text{[VB 3.41–3.45]}$$

$$k_{\mathrm{res}}(F) = k_{\mathrm{res,ref}}\frac{F}{F_{\mathrm{ref}}}, \qquad F_{\mathrm{ref}} = 1.48\times 10^{19}\ \mathrm{fiss/(m^3\,s)} \qquad \text{[VB 3.46, 3.47]}$$

The re-solved metal returns to the zone of the particles: intragranular → zone 1, grain boundary → zone 2. V. Bellani: to the single matrix.

Metal balance: one matrix → two zones

V. Bellani, continuous form [VB 3.5–3.7] and backward-Euler 3×3 system, $\mathbf{C} = (C_m, C^{\mathrm{intra}}, C^{\mathrm{GB}})$ [VB 4.15–4.18]:

$$\frac{dC_m}{dt} = y_{5\mathrm{MP}} F - J_{\mathrm{nucl,intra}} - J_{\mathrm{nucl,GB}} - (k_{\mathrm{intra}} + k_{\mathrm{GB}})(C_m - C_{\mathrm{eq}}) + k_{\mathrm{res}}(C^{\mathrm{intra}} + C^{\mathrm{GB}}) \qquad \text{[VB 3.5]}$$

$$\begin{pmatrix} 1 + (k_{\mathrm{intra}} + k_{\mathrm{GB}})\Delta t & -k_{\mathrm{res}}\Delta t & -k_{\mathrm{res}}\Delta t \\ -k_{\mathrm{intra}}\Delta t & 1 + k_{\mathrm{res}}\Delta t & 0 \\ -k_{\mathrm{GB}}\Delta t & 0 & 1 + k_{\mathrm{res}}\Delta t \end{pmatrix} \mathbf{C}^{n+1} = \begin{pmatrix} C_m^* + (k_{\mathrm{intra}} + k_{\mathrm{GB}}) C_{\mathrm{eq}} \Delta t \\ C^{\mathrm{intra},*} - k_{\mathrm{intra}} C_{\mathrm{eq}} \Delta t \\ C^{\mathrm{GB},*} - k_{\mathrm{GB}} C_{\mathrm{eq}} \Delta t \end{pmatrix} \qquad \text{[VB 4.17, 4.18]}$$

Now, after production and nucleation, $\mathbf{C} = (C_{m1}, C_{m2}, C^{\mathrm{intra}}, C^{\mathrm{GB}})$:

$$\frac{dC_{m1}}{dt} = -k_{\mathrm{intra}}(c_1 - C_{\mathrm{eq}}) - K_{\mathrm{ex}}(c_1 - c_2) + k_{\mathrm{res}} C^{\mathrm{intra}} \qquad \textbf{EC Nov. 2026}\text{, in place of [VB 3.5]}$$

$$\frac{dC_{m2}}{dt} = -k_{\mathrm{GB}}(c_2 - C_{\mathrm{eq}}) + K_{\mathrm{ex}}(c_1 - c_2) + k_{\mathrm{res}} C^{\mathrm{GB}} \qquad \textbf{EC Nov. 2026}$$

$$\frac{dC^{\mathrm{intra}}}{dt} = k_{\mathrm{intra}}(c_1 - C_{\mathrm{eq}}) - k_{\mathrm{res}} C^{\mathrm{intra}}, \qquad \frac{dC^{\mathrm{GB}}}{dt} = k_{\mathrm{GB}}(c_2 - C_{\mathrm{eq}}) - k_{\mathrm{res}} C^{\mathrm{GB}}$$

EC Nov. 2026, in place of [VB 3.6, 3.7]

production (P_k) and nucleation are applied before, as in [VB 4.2, 4.14].

Implicit 4×4 system

$\kappa_1 = k_{\text{intra}}$ if $c_1^* > C_{\text{eq}}$ else 0, $\kappa_2 = k_{\text{GB}}$ if $c_2^* > C_{\text{eq}}$ else 0 (* = after production and nucleation):

$$\begin{pmatrix} 1 + \frac{(\kappa_1 + K_{\text{ex}})\Delta t}{\phi_1} & -\frac{K_{\text{ex}}\Delta t}{\phi_2} & -k_{\text{res}}\Delta t & 0 \\ -\frac{K_{\text{ex}}\Delta t}{\phi_1} & 1 + \frac{(\kappa_2 + K_{\text{ex}})\Delta t}{\phi_2} & 0 & -k_{\text{res}}\Delta t \\ -\frac{\kappa_1 \Delta t}{\phi_1} & 0 & 1 + k_{\text{res}}\Delta t & 0 \\ 0 & -\frac{\kappa_2 \Delta t}{\phi_2} & 0 & 1 + k_{\text{res}}\Delta t \end{pmatrix} \begin{pmatrix} C_{m1} \\ C_{m2} \\ C^{\text{intra}} \\ C^{\text{GB}} \end{pmatrix}^{n+1} = \begin{pmatrix} C_{m1}^* + \kappa_1 C_{\text{eq}} \Delta t \\ C_{m2}^* + \kappa_2 C_{\text{eq}} \Delta t \\ C^{\text{intra},*} - \kappa_1 C_{\text{eq}} \Delta t \\ C^{\text{GB},*} - \kappa_2 C_{\text{eq}} \Delta t \end{pmatrix}$$

EC Nov. 2026, in place of [VB 4.17, 4.18]

- $1/\phi_k$: the fluxes depend on the local concentrations $c_k = C_{mk}/\phi_k$
- growth switched on per zone (V. Bellani: both on if $C_m^* > C_{\text{eq}}$ [VB 4.19]); re-solution always active

Particle radius

Mean spherical-equivalent radius from the precipitated inventory and the number density

$$n_j = \frac{C^j}{N_j}, \qquad R_j = \left(\frac{3 n_j \Omega_{5\mathrm{MP}}}{4\pi}\right)^{1/3} \qquad \text{[VB 3.48, 3.49]}$$

$$\Omega_{5\mathrm{MP}} = \frac{V_w}{\eta_{\mathrm{hcp}}} = \frac{1.06651 \times 10^{-29}}{0.74} = 1.44123 \times 10^{-29}\ \mathrm{m^3/at} \qquad \text{[VB 3.50–3.52]}$$

(V_w: volumes of the atomic spheres, weights 40Mo:30Ru:10Tc:15Pd:5Rh, [VB Tab. 3.2]).

Now, only the metallic part of the inventory (with OpenCalphad), $R_j = 0$ while $N_j < 1\ \mathrm{m^{-3}}$:

$$R_j = \left(\frac{3\,\phi_{\mathrm{met}}\, C^j\, \Omega_{5\mathrm{MP}}}{4\pi N_j}\right)^{1/3}, \qquad \phi_{\mathrm{met}} = 1 - x_{\mathrm{Mo}}\, f_{\mathrm{ox}} \qquad \textbf{EC Nov. 2026}\text{, in place of [VB 3.49]}$$

$\phi_{\mathrm{met}} = 1$ without thermochemistry; f_{ox} of the previous time step (next slide). The radius is used in k_j at the next step: $N_j, C^j \to R_j \to k_j \to C^j$ [VB Sec. 3.6].

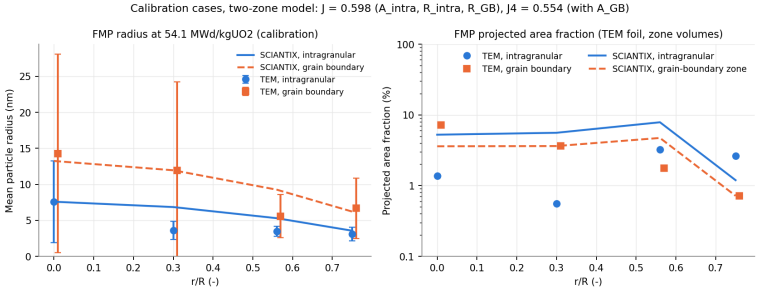
Calibration cases: UO_2, PWR, 54.1 MWd/kgUO_2

Gravelines 5 PWR standard UO_2, five cycles, section burnup 61.4 GWd/tU, TEM data of Angleraud (2024) at $r/R = 0$, 0.30, 0.56, 0.75 [VB Sec. 5.1, Tab. 5.1, 5.2]. Histories reconstructed by V. Bellani [VB Sec. 5.4, Tab. 5.6]:

Cycle	Duration (h)	F (fiss/m^{3}s)	T_0 (K)	$T_{0.30}$ (K)	$T_{0.56}$ (K)	$T_{0.75}$ (K)
1	6000	1.48×10^{19}	1171	1123	1003	869
2	6000	1.60×10^{19}	1210	1158	1029	886
3	6000	1.52×10^{19}	1181	1132	1009	873
4	6000	1.31×10^{19}	1111	1068	961	842
5	6000	1.32×10^{19}	1113	1070	963	843

- same fission rate at the four positions (scaled by $\alpha = 1.132$ to 54.1 MWd/kgUO_2 [VB 5.19–5.22]); temperature from the ALCYONE profile of the last cycle scaled with the linear power [VB 5.10–5.14]
- settings of the cases: grain radius 12.3 μm, 4.0% U-235, no He, no HBS

Calibration cases: results



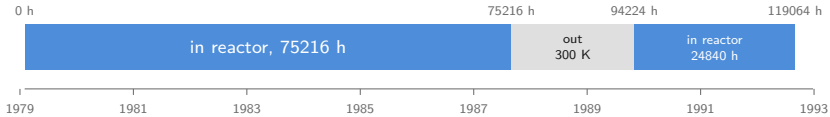
Independent validation: Buck et al. (2015), ATM-109

E.C. Buck, E.J. Mausolf, B.K. McNamara, C.Z. Soderquist, J.M. Schwantes, *Nanostructure of metallic particles in light water reactor used nuclear fuel*, J. Nucl. Mater. 461 (2015) 236–243. Not used in the calibration.

Information	Where in the paper
ATM-109, Quad Cities I BWR, GE UO_2, 3.0% U-235, initial grain size 30 μm	Sec. 1, p. 237 (Wolf et al. [21])
in reactor Feb 1979 – Sep 1987 and Nov 1989 – Sep 1992; specimens maximum burnup 70.80 GWd/MTU	Sec. 1, p. 237
cooling $\sim$21 years (July 2013); residues of nitric acid and carbonate-peroxide dissolutions	Sec. 2.1, p. 237
LWR centreline below 1200 °C, rim 400–500 °C	Sec. 1, p. 237
residue composition 20Mo:0.5–5Tc:50–60Ru:10Rh:10Pd (Adachi et al.)	Sec. 1, p. 237
SEM/TEM of the residues, hcp ε-Ru, crystallites of 1–3 nm	Secs. 2.2, 3, pp. 238–240
round particles of 10–300 nm in diameter	abstract, p. 236; Sec. 5, p. 242
cumulative *number* distribution of the particle volume, log-normal fits	Fig. 6, Sec. 3, p. 240
mean particle size 43 nm (carbonate-peroxide), 58 nm (nitric)	Sec. 3, p. 240

No power or temperature history in the paper (it would need the ATM-109 reports, Guenther et al. [10], Wolf et al. [21]) and no radial position (particles from dissolved fuel) $\Rightarrow$ assumed history, next slide.

ATM-109: irradiation history (assumed)



- continuous operation in the two periods of the paper (p. 237), 100056 h in reactor; between them $F = 0$, 300 K
- constant $F = 5.003 \times 10^{18}$ fiss/(m^3 s) $\Rightarrow$ 62.41 MWd/kgUO$_2$ = 70.8 GWd/tHM at the end: 70.80 GWd/MTU is the *maximum* burnup of the specimens, so an upper bound
- four rings of equal volume, $(r/R)^2 = 0.125$, 0.375, 0.625, 0.875, uniform F (no rim peak)
- parabolic profile $T(r) = T_s + (T_c - T_s)[1 - (r/R)^2]$, $T_c = 900\,^\circ$C (below the 1200 °C of p. 237), $T_s = 450\,^\circ$C (rim 400–500 °C, p. 237), constant in time
- grain radius 15 μm (grain size 30 μm), 3.0% U-235; other settings as the calibration cases

Case	r/R	T (K)	T (°C)
Buck2015_r035	0.354	1117	844
Buck2015_r061	0.612	1004	731
Buck2015_r079	0.791	892	619
Buck2015_r094	0.935	779	506

cases `test_FMP_UO2_Buck2015_*`
regression group `--metallicinclusions`

ATM-109: how the model is compared with Fig. 6

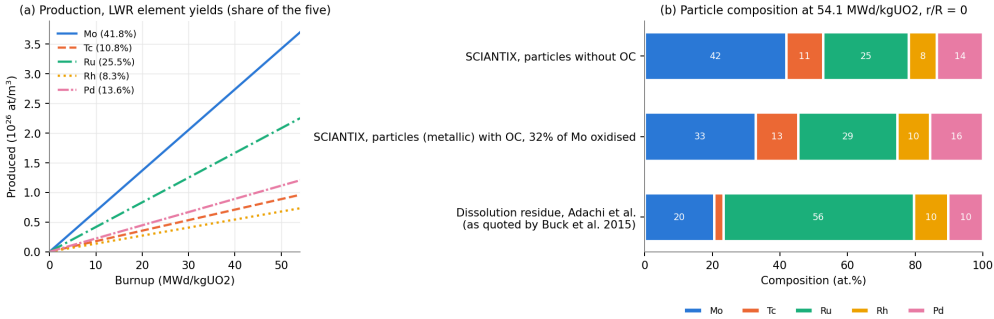
- Fig. 6 (p. 240): cumulative number fraction vs $\ln V$ (V in nm^3) for the two dissolutions; digitized (`data/buck2015/fig6_*.txt`, binned marker pixels, approximate)
- log-normal fit of $\ln V$ (`data/buck2015/summary.txt`); diameter of the equivalent sphere $d = (6V/\pi)^{1/3}$:

$$\ln d \sim \mathcal{N}(\mu_d, \sigma_d), \quad \mu_d = \frac{\mu_{\ln V} + \ln(6/\pi)}{3}, \quad \sigma_d = \frac{\sigma_{\ln V}}{3}, \quad d_{\mathrm{median}} = e^{\mu_d}, \quad \bar{d} = e^{\mu_d + \sigma_d^2/2}$$

- the fits give number-mean diameters of 50 and 62 nm, close to the 43 and 58 nm of the text: the reported values look like number means
- model: pellet-averaged population, the four rings with equal volume, intragranular and grain-boundary particles weighted by number (N_{intra}, N_{GB}); one size per ring and population

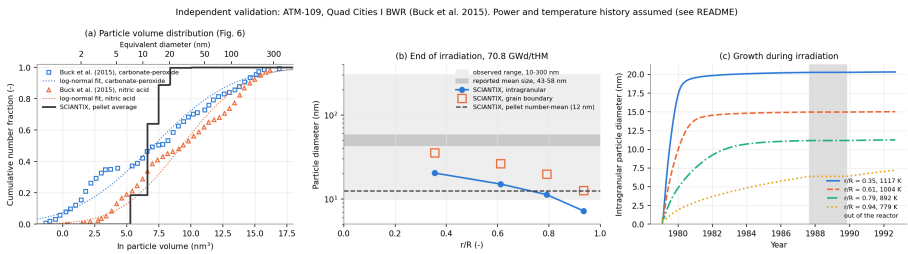
	$\mu_{\ln V}$, $\sigma_{\ln V}$	median d (nm)	number-mean d (nm)
carbonate-peroxide	7.07, 4.90	13	50 (reported 43)
nitric acid	9.30, 3.83	28	62 (reported 58)

Elements: production and particle composition



(a) Production of the five systems, LWR yields: 41.8% Mo, 10.8% Tc, 25.5% Ru, 8.3% Rh, 13.6% Pd.
(b) Particle composition = production composition; with OpenCalphad 32% of the Mo is oxidised (r000) and the metallic particles are 33Mo-13Tc-29Ru-10Rh-16Pd. Dissolution residue quoted by Buck et al. (2015, p. 237): 20Mo-0.5/5Tc-50/60Ru-10Rh-10Pd.

ATM-109: results



Ring	d_{intra} (nm)	d_{GB} (nm)	metal dissolved	metal in GB part.
0.35, 1117 K	20.3	35.5	8%	2.32%
0.61, 1004 K	15.0	26.2	14%	1.90%
0.79, 892 K	11.3	19.7	34%	1.39%
0.94, 779 K	7.2	12.6	93%	0.16%

	median d (nm)	mean d (nm)	mean ln V (V in nm^3)
Buck, carbonate-peroxide	13	50 (43)	7.07
Buck, nitric acid	28	62 (58)	9.30
SCIANTIX, two zones	11	12.5	6.78
SCIANTIX, VB parameters	19	18.5	7.95

Median just below the carbonate-peroxide series; number mean 3.5–5 times too low; distribution much narrower (one size per ring and population), the 100–300 nm particles are missing; GB particles (26–36 nm in the inner rings) closer, but 2% of the particles.

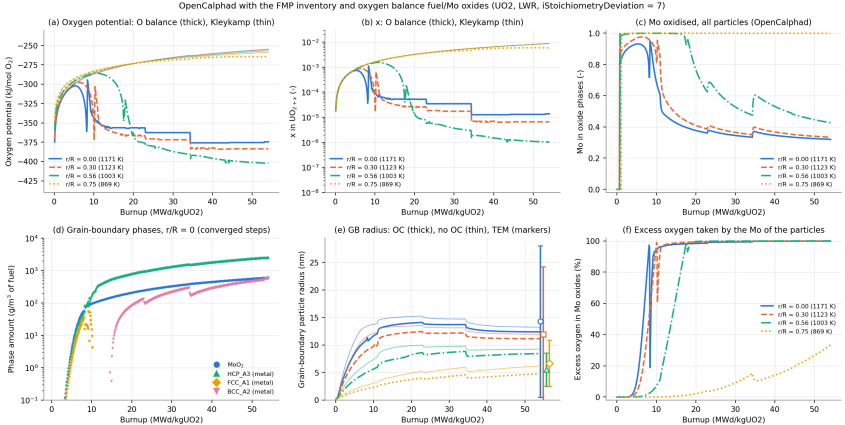
Thermochemistry cases: setup and results

`thermochemistry/test_FMP_UO2_OC_*`: the calibration cases with `iThermochimica = 2` (OpenCalphad at every step), `iStoichiometryDeviation = 7` (Kleykamp 1979, same x(burnup) at the four positions), Blackburn oxygen potential, $p = 10^5$ Pa. Regression group `--metallicinclusions-oc --oc`.

r/R	μ_{O_2} (kJ/mol)	x	Mo oxidised	R_{intra} (nm) OC / no OC	R_{GB} (nm) OC / no OC	R_{GB} (nm) TEM
0.00	−375	1.4×10^{-5}	32%	7.1 / 7.6	12.4 / 13.2	14.3
0.30	−384	6.6×10^{-6}	33%	6.4 / 6.8	11.2 / 12.0	12.0
0.56	−402	1.0×10^{-6}	43%	4.8 / 5.3	8.5 / 9.3	5.6
0.75	−265	5.8×10^{-3}	100%	2.7 / 3.6	4.8 / 6.2	6.7

- $r/R = 0$–0.56: fuel buffered at Mo/MoO_2, x far below Kleykamp (8.7×10^{-3}, −255 kJ/mol, all Mo oxidised without the balance)
- $r/R = 0.75$: 89% of the metal dissolved, the particle Mo is not enough, all oxidised; x rises with burnup
- OpenCalphad: MoO_2, HCP_A3 and BCC_A2 (metal), Mo_4O_{11} at some steps; at a few steps no convergence (previous f_{ox} kept); in r000 the convergence of one step can change between runs (final values within 10^{-5})

Thermochemistry cases: evolution



(a, b) oxygen potential and x with the oxygen balance (thick) and Kleykamp only (thin); (c) Mo oxidised; (d) phases at the grain boundary; (e) R_{GB} with and without OpenCalphad vs TEM; (f) share of the excess oxygen in the Mo oxides.

Verification, 4 November 2026

- **Regression** (`python3 -m testing.runner --metallicinclusions --metallicinclusions-oc --oc`): **12 passed, 0 failed** — 4 calibration cases, 4 ATM-109 rings, 4 thermochemistry cases; build up to date with the sources
- **Calibration still valid**: model of the calibration copy = current code (only the 4 parameter lines differ); $J_4 = 0.5541$ recomputed with the parameters in the code; $\pm 5\%$ on each parameter increases J_4 (local minimum)
- **Mass conservation**: $C_{m1} + C_{m2} + C^{\mathrm{intra}} + C^{\mathrm{GB}} = \sum_X C_X^{\mathrm{prod}}$ within 4×10^{-7} (output precision) in r000, r075, ATM-109 r035
- **Exchange**: k_{GB}^2 analytical vs finite volumes within 6×10^{-8} (`check_exchange.py`)
- all the changes tagged `EC November 2026` in the code; documentation in `validation/metallicinclusions/README.md`

Known limits and open points

- $f_{A,\mathrm{intra}}$ 4–10 times TEM at $r/R = 0$–0.30, 2 times too low at 0.75: the intragranular number density is the main remaining discrepancy
- R_{GB} too large at $r/R = 0.56$ (9.3 vs 5.6 nm)
- at low temperature most of the metal stays dissolved (89% at $r/R = 0.75$, 93% in the outer ATM-109 ring)
- one mean size per population: no size distribution; ATM-109 mean size underestimated (12.5 vs 43–58 nm)
- K_{ex} [TBC]: zone 2 as a well-mixed layer, D_{intra} also in zone 2
- **inputs inherited from V. Bellani**: grain radius 12.3 μm and 4.0% U-235, while thesis Tab. 5.1 reports a mean grain size of 8.8 μm and 4.48% U-235 (a enters ϕ_2, $N_{\mathrm{site,GB}}$, K_{ex}: a change needs a new calibration)
- calibration without OpenCalphad (the Mo oxidation reduces the radii by 5–25%)
- oxygen balance: only the particle Mo takes oxygen, excess oxygen from Kleykamp, no radial O/M
- lumped species: particle composition = production composition (measured particles poorer in Mo, richer in Ru)
- MOX systems keep fast-reactor yields: validated only for UO_2 in LWR conditions

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